

Project 2: SOC- EDR Grid

1. Project Overview

This project builds a **Security Operations Center (SOC) monitoring environment** using **Wazuh**. The objective is to detect, monitor, and analyze security events from endpoints in real time.

The system follows a **centralized EDR + SIEM architecture** where all endpoint machines send their logs to a central server for analysis.

The log flow is:

Kali Linux (Attacker)

↓

Ubuntu Desktop (Victim)

Wazuh Agent

↓

Ubuntu Server

Wazuh Manager + Filebeat + Indexer

↓

Wazuh Dashboard

2. SOC (Security Operations Center)

A **Security Operations Center (SOC)** is a centralized team or system responsible for monitoring and defending an organization's IT infrastructure.

Main responsibilities of SOC include:

- Continuous monitoring of systems (24×7)
- Detecting cyber attacks in real time
- Investigating suspicious alerts
- Responding to security incidents
- Maintaining system security posture

The SOC uses tools like **Wazuh SIEM** to analyze security logs and identify threats.

3. EDR (Endpoint Detection and Response)

EDR (Endpoint Detection and Response) focuses on monitoring endpoint devices such as:

- Servers
- Workstations
- Virtual Machines
- Cloud instances

The purpose of EDR is to detect suspicious behavior occurring directly on the endpoint.

In this project, **Wazuh Agent acts as the EDR component.**

The agent collects:

- System logs
- Authentication logs
- File integrity changes
- Malware activity
- Unauthorized access attempts

These logs are then sent to the Wazuh Manager for analysis

4. Ubuntu Server – Wazuh Manager (SOC Server)

The Ubuntu Server acts as the central brain of the SOC environment.

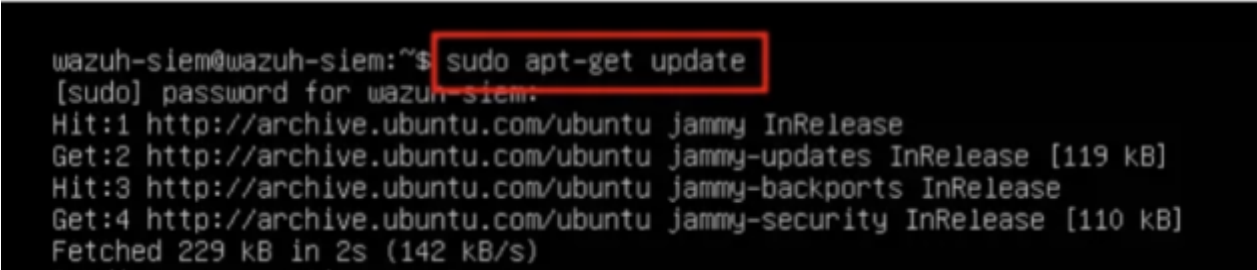
Components installed on the server include:

- Wazuh Manager: Without the Wazuh Manager, agents cannot communicate or send logs.
- Wazuh Indexer: The Wazuh Indexer is responsible for storing security logs.
- Filebeat: Filebeat is a log forwarding tool. Enable data visualization on the dashboard
- Wazuh Dashboard

Conclusion:

- Endpoint monitoring using Wazuh Agent
- Centralized log analysis using Wazuh Manager
- Security event visualization through the Wazuh Dashboard
- Real-time detection of potential threats

Figure 1: Installation process



```
wazuh-siem@wazuh-siem:~$ sudo apt-get update
[sudo] password for wazuh-siem:
Hit:1 http://archive.ubuntu.com/ubuntu jammy InRelease
Get:2 http://archive.ubuntu.com/ubuntu jammy-updates InRelease [119 kB]
Hit:3 http://archive.ubuntu.com/ubuntu jammy-backports InRelease
Get:4 http://archive.ubuntu.com/ubuntu jammy-security InRelease [110 kB]
Fetched 229 kB in 2s (142 kB/s)
```

Figure 2: setup of Wazuh

```
root@wazuh-siem:/home/wazuh-siem# apt install curl
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
curl is already the newest version (7.81.0-1ubuntu1.15).
curl set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 47 not upgraded.
root@wazuh-siem:/home/wazuh-siem# curl -sO https://packages.wazuh.com/4.7/wazuh-install.sh && sudo b
ash ./wazuh-install.sh -a
14/01/2024 13:34:51 INFO: Starting Wazuh installation assistant. Wazuh version: 4.7.2
14/01/2024 13:34:51 INFO: Verbose logging redirected to /var/log/wazuh-install.log
14/01/2024 13:35:00 INFO: Wazuh web interface port will be 443.
14/01/2024 13:35:06 INFO: --- Dependencies ---
14/01/2024 13:35:06 INFO: Installing apt-transport-https.
14/01/2024 13:35:18 INFO: Wazuh repository added.
14/01/2024 13:35:18 INFO: --- Configuration files ---
14/01/2024 13:35:18 INFO: Generating configuration files.
14/01/2024 13:35:20 INFO: Created wazuh-install-files.tar. It contains the Wazuh cluster key, certif
icates, and passwords necessary for installation.
14/01/2024 13:35:20 INFO: --- Wazuh indexer ---
14/01/2024 13:35:20 INFO: Starting Wazuh indexer installation.
14/01/2024 13:38:05 INFO: Wazuh indexer installation finished.
14/01/2024 13:38:06 INFO: Wazuh indexer post-install configuration finished.
14/01/2024 13:38:06 INFO: Starting service wazuh-indexer.
14/01/2024 13:38:31 INFO: wazuh-indexer service started.
14/01/2024 13:38:31 INFO: Initializing Wazuh indexer cluster security settings.
14/01/2024 13:38:42 INFO: Wazuh indexer cluster initialized.
14/01/2024 13:38:42 INFO: --- Wazuh server ---
14/01/2024 13:38:42 INFO: Starting the Wazuh manager installation.
14/01/2024 13:40:08 INFO: Wazuh manager installation finished.
14/01/2024 13:40:08 INFO: Starting service wazuh-manager.
-
```

- Used to install and connect the endpoint (Ubuntu Desktop) to the Wazuh Manager server.

Figure 3: configure the Wazuh Agent on Ubuntu Desktop

```
user@ubuntu-desktop: ~
user@ubuntu-desktop:~$ curl -sO https://packages.wazuh.com/4.x/wazuh-agent_4.7.5-1_amd64.deb
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Speed
%   Tond      Upload    Total Upload  Total  Spent    Left   Speed
100% 622k  622k   515 0    515k  --:--  --:--  --:--  515k
wazuh-agent_4.7.5-1_amd64.deb

user@ubuntu-desktop:~$ sudo WAZUH_MANAGER='192.168.64.3' dpkg -i wazuh-agent_4.7.5-1_amd64.deb
(Reading database ... 179917 files and directories currently installed.)
Preparing to unpack wazuh-agent_4.7.5-1_amd64.deb ...
Unpacking wazuh-agent (4.7.5-1) over (4.7.5-1) ...
Setting up wazuh-agent (4.7.5-1) ...
Registered "Wazuh agent 4.7.5-1". [loaded plugin 'oscap'.
Started wazuh-modulesd...
Enabled all Wazuh agent modules if they're a roots manager in the manager's configuration.
Started wazuh-execd...
INFO: (1217): Registering agent to manager (192.168.64.3).

user@ubuntu-desktop:~$ sudo systemctl daemon-reload
sudo systemctl daemon-reload sudo systemctl enable wazuh-agent
sudo systemctl start wazuh-agent since Tue 2024-04-23 16:38:32 PDT; 1s ago
sudo systemctl status wazuh-agent

user@ubuntu-desktop:~$
```

- The Wazuh agent is installed on the Ubuntu Desktop endpoint to collect security logs and send them to the Wazuh Manager. This enables centralized monitoring, threat detection, and incident response in the SOC environment.

Figure 4: Certificates creation

```
root@wazuh-siem:/home/wazuh-siem/wazuh-installer# curl -sO https://packages.wazuh.com/4.7/wazuh-certs-tool.sh
root@wazuh-siem:/home/wazuh-siem/wazuh-installer# curl -sO https://packages.wazuh.com/4.7/config.yml
root@wazuh-siem:/home/wazuh-siem/wazuh-installer# ls
config.yml wazuh-certs-tool.sh
root@wazuh-siem:/home/wazuh-siem/wazuh-installer#
```

- Certificates are using for **encryption** the content.

Figure 5: `ip` a command to retrieve your server's IP

```
root@wazuh-siem:/home/wazuh-siem/wazuh-installer# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:8c:1d:0b brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.251.150/24 metric 100 brd 192.168.251.255 scope global dynamic ens33
        valid_lft 1010sec preferred_lft 1010sec
    inet6 fe80::20c:29ff:fe8c:1d0b/64 scope link
        valid_lft forever preferred_lft forever
3: docker0: <NO-CARRIER,BROADCAST,MULTICAST,UP> mtu 1500 qdisc noqueue state DOWN group default
    link/ether 02:42:f8:c9:34:37 brd ff:ff:ff:ff:ff:ff
    inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
        valid_lft forever preferred_lft forever
6: vxlan.calico: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1450 qdisc noqueue state UNKNOWN group default
    link/ether 66:06:41:a0:3b:ce brd ff:ff:ff:ff:ff:ff
    inet 10.1.26.64/32 scope global vxlan.calico
        valid_lft forever preferred_lft forever
    inet6 fe80::6406:41ff:fea0:3bce/64 scope link
        valid_lft forever preferred_lft forever
7: calid6fabdc3875@if3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1450 qdisc noqueue state UP group default
    link/ether ee:ee:ee:ee:ee:ee brd ff:ff:ff:ff:ff:ff link-netns cni-d61607d5-aa2a-161d-86bb-50a27eae12b
    inet6 fe80::ecee:eeff:feee:eeee/64 scope link
        valid_lft forever preferred_lft forever
8: calic54cd419f2@if3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1450 qdisc noqueue state UP group default
    link/ether ee:ee:ee:ee:ee:ee brd ff:ff:ff:ff:ff:ff link-netns cni-5e65c97a-698f-41ce-6820-75cf85c03854
    inet6 fe80::ecee:eeff:feee:eeee/64 scope link
        valid_lft forever preferred_lft forever
root@wazuh-siem:/home/wazuh-siem/wazuh-installer#
```

```
GNU nano 6.2
nodes:
# Wazuh indexer nodes
indexer:
- name: node-1
  ip: "192.168.251.150"
#- name: node-2
# ip: "<indexer-node-ip>"
#- name: node-3
# ip: "<indexer-node-ip>"

# Wazuh server nodes
# If there is more than one Wazuh server
# node, each one must have a node_type
server:
- name: wazuh-1
  ip: "192.168.251.150"
# node_type: master
#- name: wazuh-2
# ip: "<wazuh-manager-ip>"
# node_type: worker
#- name: wazuh-3
# ip: "<wazuh-manager-ip>"
# node_type: worker

# Wazuh dashboard nodes
dashboard:
- name: dashboard
  ip: "192.168.251.150"
```

Figure 6:Installing Wazuh indexer.

```
root@wazuh-siem:/home/wazuh-siem/wazuh-installer# apt-get -y install wazuh-indexer
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  wazuh-indexer
0 upgraded, 1 newly installed, 0 to remove and 47 not upgraded.
Need to get 686 MB of archives.
After this operation, 969 MB of additional disk space will be used.
Get:1 https://packages.wazuh.com/4.x/apt/stable/main amd64 wazuh-indexer amd64 4.7.2-1 [686 MB]
Fetched 686 MB in 1min 35s (7,216 kB/s)
Selecting previously unselected package wazuh-indexer.
(Reading database ... 74412 files and directories currently installed.)
Preparing to unpack .../wazuh-indexer_4.7.2-1_amd64.deb ...
Creating wazuh-indexer group... OK
Creating wazuh-indexer user... OK
Unpacking wazuh-indexer (4.7.2-1) ...
Setting up wazuh-indexer (4.7.2-1) ...
Created opensearch keystore in /etc/wazuh-indexer/opensearch.keystore
Processing triggers for libc-bin (2.35-0ubuntu3.6) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@wazuh-siem:/home/wazuh-siem/wazuh-installer#
```

Figure 7:Configuring Wazuh indexer.

```
network.host: 0.0.0.0
node.name: "node-1"
cluster.initial_master_nodes:
  - "node-1"
#- "node-2"
#- "node-3"
cluster.name: "wazuh-cluster"
discovery.seed_hosts:
  # - "node-1-ip"
  # - "node-2-ip"
  # - "node-3-ip"
node.max_local_storage_nodes: ""
path.data: /var/lib/wazuh-indexer
path.logs: /var/log/wazuh-indexer

plugins.security.ssl.transport.pemcert_filepath: /etc/wazuh-indexer/certs/indexer.pem
plugins.security.ssl.transport.pemkey_filepath: /etc/wazuh-indexer/certs/indexer-key.pem
plugins.security.ssl.transport.trustedcas_filepath: /etc/wazuh-indexer/certs/root-ca.pem
plugins.security.ssl.transport.pemcert_filepath: /etc/wazuh-indexer/certs/indexer.pem
plugins.security.ssl.transport.pemkey_filepath: /etc/wazuh-indexer/certs/indexer-key.pem
plugins.security.ssl.transport.trustedcas_filepath: /etc/wazuh-indexer/certs/root-ca.pem
plugins.security.ssl.transport.enabled: true
plugins.security.ssl.transport.enforce_hostname_verification: false
plugins.security.ssl.transport.resolve_hostname: false

plugins.security.authz.admin_dn:
  - "CN=admin,O=Wazuh,C=US"
plugins.security.check_snapshot_restore_write_privileges: true
plugins.security.enable_snapshot_restore_privileges: true
plugins.security.nodes_dn:
  - "CN=node-1,O=Wazuh,C=US"
  - "CN=node-2,O=Wazuh,C=US"
  - "CN=node-3,O=Wazuh,C=US"
plugins.security.restapi.roles_enabled:
  - "all_access"
  - "security_rest_api_access"

plugins.security.system_indices.enabled: true
plugins.security.system_indices.indices: ["plugin-el-model", "plugin-el-task", "opendistro-alerting-config", "opendistro-alerting-alerts", "opendistro-anomaly-results", "opendistro-anomaly"]

## Option to allow Filebeat-es 7.10.2 to work ##
compatibility.override_main_response_version: true
```


Figure 8: Running Status

```
root@wazuh-siem:/home/wazuh-siem/wazuh-installer# systemctl status wazuh-indexer
● wazuh-indexer.service - Wazuh-indexer
   Loaded: loaded (/lib/systemd/system/wazuh-indexer.service; enabled; vendor preset: enabled)
   Active: active (running) since Sun 2024-01-14 16:45:08 UTC; 26s is ago
     Docs: https://documentation.wazuh.com
    Main PID: 117618 (java)
      Tasks: 50 (limit: 4515)
     Memory: 1.3G
        CPU: 54.318s
    CGroup: /system.slice/wazuh-indexer.service
            └─117618 /usr/share/wazuh-indexer/jdk/bin/java -Xshare:auto -Dopensearch.networkaddress.cache.ttl=60 -Dopensearch.networkaddress.cache.negative.ttl=10 -XX:+AlwaysPreTouch -Xss10m -Djv...
```

Figure 9:Installing Filebeat.

```
root@wazuh-siem:/home/wazuh-siem/wazuh-installer# apt-get -y install filebeat
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  filebeat
0 upgraded, 1 newly installed, 0 to remove and 47 not upgraded.
Need to get 22.1 MB of archives.
After this operation, 73.6 MB of additional disk space will be used.
Get:1 https://packages.wazuh.com/4.x/apt/stable/main amd64 filebeat amd64 7.10.2 [22.1 MB]
Fetched 22.1 MB in 4s (5,371 kB/s)
Selecting previously unselected package filebeat.
(Reading database ... 96884 files and directories currently installed.)
Preparing to unpack .../filebeat_7.10.2_amd64.deb ...
Unpacking filebeat (7.10.2) ...
Setting up filebeat (7.10.2) ...
Scanning processes ...
Scanning linux images ...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
root@wazuh-siem:/home/wazuh-siem/wazuh-installer#
```

Figure 10: Configuring Filebeat

```

GNU nano 6.2 /etc/Filebeat/Filebeat.yml
# This requires a Kibana endpoint configuration.
setup.kibana:

# Kibana Host
# Scheme and port can be left out and will be set to the default (http and 5601)
# In case you specify an additional path, the scheme is required: http://localhost:5601/path
# IPv6 addresses should always be defined as: https://[2001:db8::1]:5601
#host: "localhost:5601"

# Kibana Space ID
# ID of the Kibana Space into which the dashboards should be loaded. By default,
# the Default Space will be used.
#space.id:

# ----- Elastic Cloud -----
# These settings simplify using Filebeat with the Elastic Cloud (https://cloud.elastic.co/).
# The cloud.id setting overwrites the 'output.elasticsearch.hosts' and
# 'setup.kibana.host' options.
# You can find the 'cloud.id' in the Elastic Cloud web UI.
#cloud.id:

# The cloud.auth setting overwrites the 'output.elasticsearch.username' and
# 'output.elasticsearch.password' settings. The format is 'user:pass'.
#cloud.auth:

# ----- Outputs -----
# Configure what output to use when sending the data collected by the beat.

# ----- Elasticsearch Output -----
output.elasticsearch:
  # Array of hosts to connect to.
  hosts: ["192.168.251.150:9200"]
  # Protocol - either 'http' (default) or 'https'.
  protocol: "https"
  # Authentication credentials - either API key or username/password.
  #api_key: "id:api_key"
  username: $username
  password: $password

# ----- Logstash Output -----
output.logstash:
  # The Logstash Hosts
  #hosts: ["localhost:5044"]

# Optional SSL. By default is off.
# List of root certificates for HTTPS server verifications
#ssl.certificate_authorities: ["/etc/pki/root/ca.pem"]

```

Figure 11: Filebeat running status

```

root@wazuh-siem:/home/wazuh-siem/wazuh-installer# filebeat test output
elasticsearch: https://192.168.251.150:9200...
  parse url ... OK
  connection ...
    parse host ... OK
    dns lookup ... OK
    addresses: 192.168.251.150
    dial up ... OK
  TLS ...
    security: server's certificate chain verification is enabled
    handshake ... ERROR x509: certificate signed by unknown authority
root@wazuh-siem:/home/wazuh-siem/wazuh-installer#

```

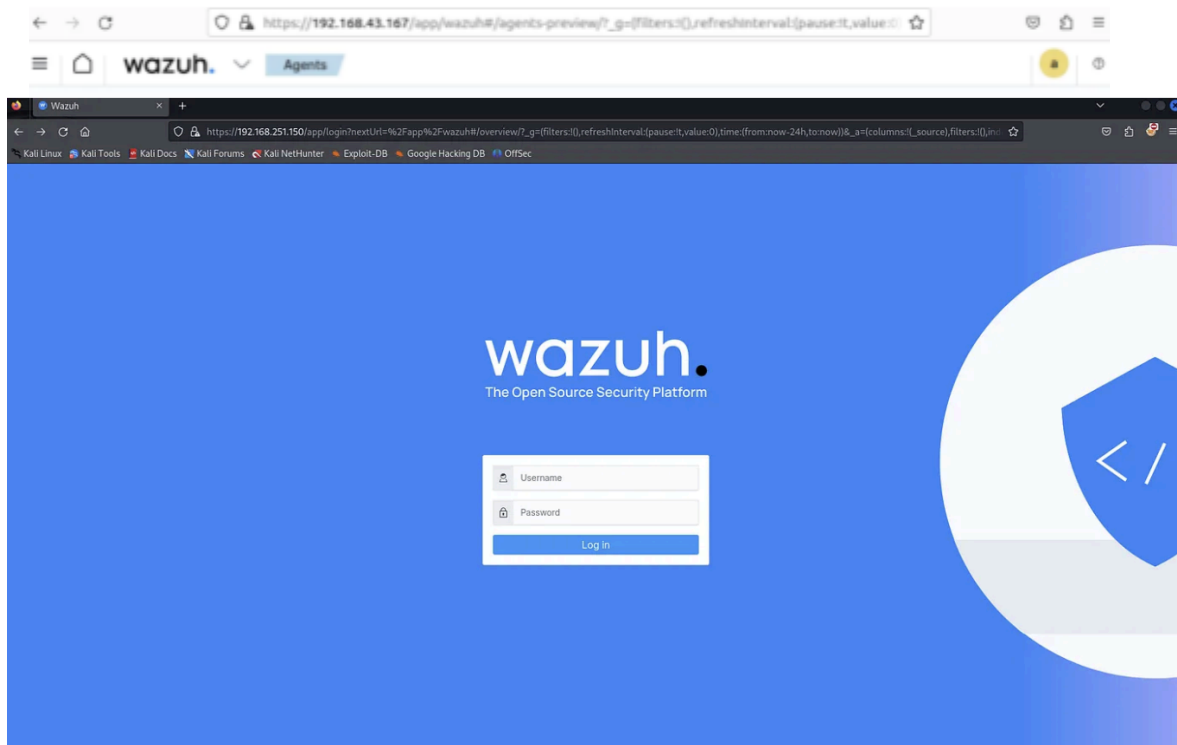
Figure 12: Wazuh Dashboard

```

GNU nano 6.2 /etc/wazuh-dashboard/opensearch_dashboards.yml
server.host: 0.0.0.0
server.port: 443
opensearch.hosts: https://192.168.251.150:9200
opensearch.ssl.verificationMode: certificate
#opensearch.username:
#opensearch.password:
opensearch.requestHeadersAllowlist: ["securitytenant","Authorization"]
opensearch_security.multitenancy.enabled: false
opensearch_security.readonly_mode.roles: ["kibana_read_only"]
server.ssl.enabled: true
server.ssl.key: "/etc/wazuh-dashboard/certs/dashboard-key.pem"
server.ssl.certificate: "/etc/wazuh-dashboard/certs/dashboard.pem"
opensearch.ssl.certificateAuthorities: ["/etc/wazuh-dashboard/certs/root-ca.pem"]
uiSettings.overrides.defaultRoute: /app/wazuh

```

Figure 13: Final Output



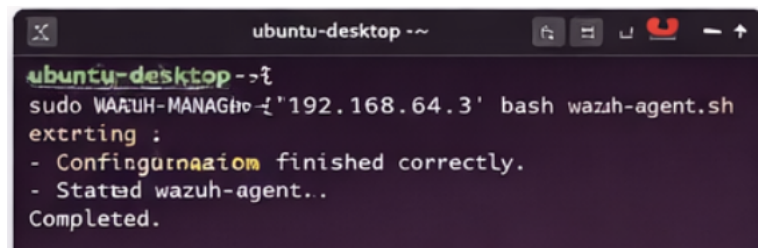
Week 3&4: Threat Detection and Security Monitoring.

- Wazuh – EDR / SIEM platform
- OpenSearch – log indexing and analytics
- Kibana – visualization dashboards
- Hydra – brute force simulation tool
- Atomic Red Team – adversary attack simulation framework

The goal of Week 3&4 is to verify that the SOC environment can **detect suspicious activities and security threats** using the Wazuh monitoring platform.

At this stage:

- The Wazuh Manager is installed
- The Wazuh Agent is connected to the endpoint
- Logs are being collected from the endpoint system

A terminal window titled 'ubuntu-desktop ~' showing the installation of the Wazuh agent. The prompt is 'ubuntu-desktop ~\$'. The command entered is 'sudo WAZUH-MANAGER -i '192.168.64.3' bash wazuh-agent.sh'. The output shows 'extrting :' followed by two lines: '- Configuration finished correctly.' and '- Started wazuh-agent...'. The final output is 'Completed.'.

```
ubuntu-desktop ~$ sudo WAZUH-MANAGER -i '192.168.64.3' bash wazuh-agent.sh
extrting :
- Configuration finished correctly.
- Started wazuh-agent...
Completed.
```

Now the SOC system begins **monitoring and analyzing security events in real time**.

2. File Integrity Monitoring (FIM)

One of the most important features of Wazuh is **File Integrity Monitoring (FIM)**. FIM monitors critical system files and detects any unauthorized changes.

Examples of monitored files:

- /etc/passwd
- /etc/shadow

```

user@ubuntu-desktop: ~
user@ubuntu-desktop:~$ sudo nano /etc/passwd
[sudo] password for user:

GNU nano 6.2 /etc/passwd

root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin::/usr/sbin:nologin
bin:x:2:2:bin:/bin::/usr/sbin:nologin
sys:x:3:3:sys:/dev::/usr/sbin:nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/bin:nologin
man:x:6:12:man:/var/cache/man:/usr/sbin:nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin:nologin
user:x:1000:1000:Ubuntu User:/home/user:/binbash

```

- /etc/sudoers

3. Log Monitoring:

This log records:

- Login attempts
- Failed authentication
- Privilege escalation attempts

Monitoring these logs helps detect attacks such as:

- brute force login attempts
- unauthorized access
- suspicious user activity

hydra -l user -P rockyou.txt ssh://192.168.64.3 (attack using this command)

- hydra → password cracking tool
- -l → username
- -P → password list
- ssh:// → target service

The screenshot shows the Wazuh dashboard interface. On the left sidebar, there are statistics for agents: Total agents (1), Active agents (1), Disconnected agents (0), Pending agents (0), and Never connected agents (0). The main panel displays a prominent yellow alert box with a red warning icon. The alert title is "Check if the /etc/passwd file has been modified". The description states: "The /etc/passwd file modified. This file contains local account and their access permissions. An unauthorized change could allow unauthorized access. Src IP: 192.168.64.20." Below the alert box, there is a table with columns: Level, Rule, Description, and Security Information. The table contains one entry with Level 7, Rule 554, and a description that matches the alert. The Security Information column lists the location as /etc/passwd and provides several hashes (MD5, SHA9, SHA8, SHA256) for the file. At the bottom, there is a log message section showing the event details.

4. Incident Investigation

- Event logs
- Source IP address
- Affected endpoint
- User activity

The screenshot displays the Wazuh dashboard interface for a detailed event view. The top navigation bar includes the Wazuh logo, a menu icon, and a user profile icon. The main content area is divided into several sections:

- Alerts:** A summary bar shows 7 alerts, with 5 MITRE, 1 fs, and 1 fim. The selected alert is Rule ID: 554, 'File /etc/passwd modified', triggered 'About a minute ago'.
- Event Details:** Provides information about the event, including the timestamp (February 28, 2026, at 19:08:03), location (root), and agent (ubuntu).
- Log message:** Displays the message 'File /etc/passwd modified.'
- Raw log:** Shows the raw log entry: 'pwd-n0Bppw FILE=/etc/passwd MODE=whodat'.
- MITRE ATT&CK:** A sidebar on the right provides details about the MITRE ATT&CK framework, including Technique ID (T1070), Technique (Indicator Removal on Host), Tactic (Defense Evasion), Mitre ID (611), and Level (7).
- Agent:** A sidebar on the right shows agent information: Agent ID (001), Name (ubuntu-desktop), Agent IP (192.168.64.20), and Status (Active).
- Related Logs:** A section at the bottom right shows related events (3 Alert) and triggered rules (2 Rule).

clear visual example of a MITRE ATT&CK mapping screen

Alert MITRE ATT&CK Mapping

Rule ID: 554
File '/etc/passwd' md' modified
Level: **7**
Agent: ubuntu-desktop

Technique ID: T1070
Tactic: Defense Evasion
MITRE ID: T1070

Technique ID	Technique
T1059	Command Scripting Interpreter
T1068	Exploitation Privilege Escalation
T1070	Indicator Removal on Host
T1003	OS Credential Dumping

T1077 Indicator Removal

Technique ID: T1070 Indicator removal on Host
Tactic: Defense Evasion
MITRE ID: T1070
Level: 7
[View techniques on the MITRE ATT&CK matrix](#)

Techniques

- T1059 Command and Scripting Interpreter
- T1068 Exploitation for Privilege Escalation
- T1070 Indicator Removal on Host
- T1003 OS Credential Dumping

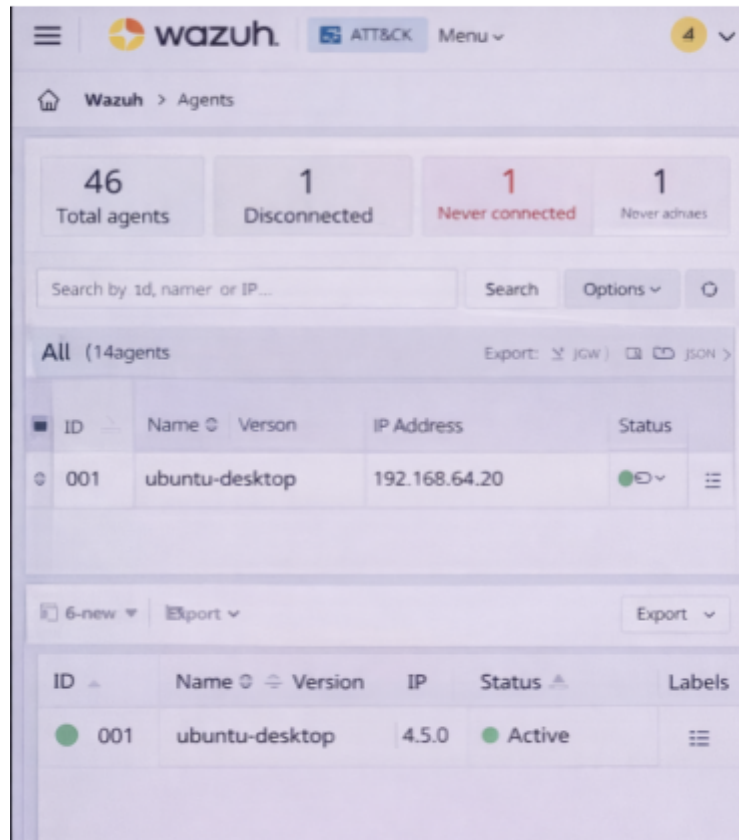
MITRE ATT&CK Mapping

- Maps alerts to attacker techniques
- shows attack stages and tactics
- helps analysts understand attacker behavior.

Agents Dashboard

- displays all monitored endpoints
- shows agent status and IP address
- enables real-time endpoint security monitoring.

Dashboard Overview



The project successfully demonstrated:

1. Detection of SSH brute force attacks.
2. Automatic blocking of attacker IP.
3. Simulation of ransomware techniques.
4. Visualization of attack lifecycle in SOC dashboard.

Screenshots collected:

- Hydra brute force attack
- Wazuh alert logs
- Firewall blocking rule
- Kibana attack visualization

Advantages

Sentient Shield provides:

- Real-time threat detection
- automated incident response
- centralized log monitoring
- threat hunting capabilities
- MITRE ATT&CK mapping

Conclusion

This project demonstrates the development of an enterprise SOC monitoring system capable of detecting and responding to cyber threats in real time. By integrating **Wazuh**, **OpenSearch**, and **Kibana**, the system provides automated security monitoring and response.

The implementation of active response and threat simulation highlights the effectiveness of the **Sentient Shield EDR Grid** in identifying and mitigating modern cyber attacks.